

Template for multipart MOM problem

Common control:

```
$anstypes = array("choices","choices")
```

```
$answeights = array(.4,.6)
```

```
// Student guide example graphs
```

```
$guide_xmin = -7
```

```
$guide_xmax = 7
```

```
$guide_ymin = -7
```

```
$guide_ymax = 7
```

```
$guide_labels = "2:2:x:y"
```

```
$guide_grid = 1
```

```
$guide_w = 165
```

```
$guide_h = 165
```

```
$guide_lin =
```

```
showplot("x+1,black,-6,6,arrow,arrow,2",$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
```

```
$guide_lin = replacealltext($guide_lin,"A non-vertical line. It is a function because each x-value has one y-value.")
```

```
$guide_vert =
```

```
showplot("x=2,black,-6,6,arrow,arrow,2",$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
```

```
$guide_vert = replacealltext($guide_vert,"A vertical line. It is not a function because one x-value has many y-values.")
```

```
$guide_quad =
```

```
showplot("(x+1)^2-4,black,-4,2,arrow,arrow,2",$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
```

```
$guide_quad = replacealltext($guide_quad,"A vertical parabola. It is a function because each x-value has one y-value.")
```

```
$guide_side =
```

```
showplot("[(t)^2-5,t],black,-3,3,arrow,arrow,2",$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
```

```
$guide_side = replacealltext($guide_side,"A sideways parabola. It is not a function because some x-values have two y-values.")
```

```
$guide_exp =
```

```
showplot("2^(x-1)-3,black,-3,4,arrow,arrow,2",$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
```

```
$guide_exp = replacealltext($guide_exp,"An exponential curve. It is a function because each x-value has one y-value.")
```

```
$guide_twoexp_funcs = array(
```

```
  "2^(x-1),black,-4,3,arrow,arrow,2",
```

```
  "-(2^(x-1)),black,-4,3,arrow,arrow,2"
```

```
)
```

```

$guide_twoexp =
showplot($guide_twoexp_funcs,$guide_xmin,$guide_xmax,$guide_ymin,$guide_ymax,$guide_labels,$guide_grid,$guide_w,$guide_h)
$guide_twoexp = replacealltext($guide_twoexp,"Two exponential-looking branches. It is not a function because many x-values have two y-values.")

// 0 = linear function
// 1 = vertical line, not a function
// 2 = quadratic function
// 3 = sideways parabola, not a function
// 4 = exponential function
// 5 = two-branch exponential relation, not a function
$graphtype = rand(0,5)

// Part A choices
$questions[0] = array("Yes", "No")

// Graph settings
$xmin = -10
$xmax = 10
$ymin = -10
$ymax = 10
$labels = "2:2:x:y"
$grid = 1
$gw = 330
$gh = 330

if ($graphtype==0) {
  // Linear function: passes vertical line test
  $m = randfrom(array(-2,-1,1,2))
  $b = rand(-2,2)

  if (abs($m)==2) {
    $xlo = -4
    $xhi = 4
  } else {
    $xlo = -7
    $xhi = 7
  }
}

$func = "$m*x+($b),black,$xlo,$xhi,arrow,arrow,2"
$graph = showplot($func,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)

$answer[0] = 0

$bchoices = array(
  "Each x-value has one y-value.",
  "Some x-values have two y-values.",
  "Curved graphs are not functions.",
  "Crossing an axis makes it fail."
)

```

```
$btags = array("correct","wrong","wrong","wrong")
```

```
$alt = "A non-vertical straight line on the coordinate plane."
```

```
$showanswer = "Yes, the graph represents a function. It passes the vertical line test because  
each x-value has exactly one y-value."
```

```
}
```

```
if ($graphtype==1) {
```

```
// Vertical line: not a function
```

```
$c = nonzerorand(-7,7)
```

```
// Parametric vertical segment with arrows
```

```
$func = "[$c,t],black,-8,8,arrow,arrow,2"
```

```
$graph = showplot($func,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)
```

```
$answer[0] = 1
```

```
$bchoices = array(
```

```
"Some x-values have two y-values.",
```

```
"Each x-value has one y-value.",
```

```
"All straight lines are functions.",
```

```
"Crossing an axis makes it fail."
```

```
)
```

```
$btags = array("correct","wrong","wrong","wrong")
```

```
$alt = "A vertical line segment on the coordinate plane."
```

```
$showanswer = "No, the graph does not represent a function. A vertical line gives the same  
x-value many different y-values, so it fails the vertical line test."
```

```
}
```

```
if ($graphtype==2) {
```

```
// Quadratic function: passes vertical line test
```

```
$a = randfrom(array(-1,1))
```

```
$h = rand(-3,3)
```

```
if ($a==1) {
```

```
$k = rand(-5,-1)
```

```
} else {
```

```
$k = rand(1,5)
```

```
}
```

```
$qmin = $h - 3
```

```
$qmax = $h + 3
```

```
$func = "($a)*(x-($h))^2+($k),black,$qmin,$qmax,arrow,arrow,2"
```

```
$graph = showplot($func,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)
```

```
$answer[0] = 0
```

```
$bchoices = array(
```

```
"Each x-value has one y-value.",
```

```
"Some x-values have two y-values.",
```

```

    "Having a vertex makes it fail.",
    "Curved graphs are not functions."
)
$btags = array("correct","wrong","wrong","wrong")

$alt = "A vertical parabola on the coordinate plane."
$showanswer = "Yes, the graph represents a function. A vertical parabola passes the vertical line
test because each x-value has exactly one y-value."
}

if ($graphtype==3) {
    // Sideways parabola: not a function
    $a = randfrom(array(-1,1))

    if ($a==1) {
        $h = rand(-6,-2)
    } else {
        $h = rand(2,6)
    }

    $k = rand(-2,2)
    $tmin = $k - 3
    $tmax = $k + 3

    //  $x = a(y-k)^2+h$ , written parametrically as  $[x(t),y(t)]$ 
    $func = "[($a)*(t-($k))^2+($h),t],black,$tmin,$tmax,arrow,arrow,2"
    $graph = showplot($func,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)

    $answer[0] = 1

    $bchoices = array(
        "Some x-values have two y-values.",
        "Each x-value has one y-value.",
        "Having a vertex makes it fail.",
        "All parabolas are functions."
    )
    $btags = array("correct","wrong","wrong","wrong")

    $alt = "A sideways parabola on the coordinate plane."
    $showanswer = "No, the graph does not represent a function. Some vertical lines would hit the
graph twice, meaning some x-values have two different y-values."
}

if ($graphtype==4) {
    // Exponential function: passes vertical line test
    $a = randfrom(array(-2,-1,1,2))
    $h = rand(-2,2)
    $k = rand(-1,1)
    $dir = rand(0,1)

    if ($dir==0) {
        // exponential growth shape

```

```

$semin = $h - 4
$semax = $h + 2
$func = "($a)*(2^(x-($h)))+($k),black,$semin,$semax,arrow,arrow,2"
} else {
  // exponential decay shape
  $semin = $h - 2
  $semax = $h + 4
  $func = "($a)*(2^(-(x-($h))))+($k),black,$semin,$semax,arrow,arrow,2"
}

$graph = showplot($func,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)

$answer[0] = 0

$bchoices = array(
  "Each x-value has one y-value.",
  "Some x-values have two y-values.",
  "Exponential graphs always fail.",
  "Curved graphs are not functions."
)
$btags = array("correct","wrong","wrong","wrong")

$salt = "An exponential curve on the coordinate plane."
$showanswer = "Yes, the graph represents a function. Even though it curves, each x-value has
exactly one y-value, so it passes the vertical line test."
}

if ($graphtype==5) {
  // Two-branch exponential relation: not a function
  // This replaces the earlier sideways exponential because that graph still passed the vertical line
  test.
  $a = randfrom(array(1,2))
  $h = rand(-2,2)
  $k = rand(-1,1)
  $dir = rand(0,1)

  if ($dir==0) {
    $semin = $h - 4
    $semax = $h + 2

    $funcs = array(
      "($a)*(2^(x-($h)))+($k),black,$semin,$semax,arrow,arrow,2",
      "-($a)*(2^(x-($h)))+($k),black,$semin,$semax,arrow,arrow,2"
    )
  } else {
    $semin = $h - 2
    $semax = $h + 4

    $funcs = array(
      "($a)*(2^(-(x-($h))))+($k),black,$semin,$semax,arrow,arrow,2",
      "-($a)*(2^(-(x-($h))))+($k),black,$semin,$semax,arrow,arrow,2"
    )
  }
}

```

```

}

$graph = showplot($funcs,$xmin,$xmax,$ymin,$ymax,$labels,$grid,$gw,$gh)

$answer[0] = 1

$bchoices = array(
  "Some x-values have two y-values.",
  "Each x-value has one y-value.",
  "All exponential graphs are functions.",
  "Crossing an axis makes it fail."
)
$btags = array("correct","wrong","wrong","wrong")

$alt = "Two exponential-looking branches on the coordinate plane, one above and one below."
$showanswer = "No, the graph does not represent a function. Many vertical lines would hit both
branches, so some x-values have two different y-values."
}

// Accessibility
$graph = replacealttext($graph,$alt)

// Shuffle Part B choices while keeping the correct tag paired with the correct choice.
$bchoices,$btags = jointshuffle($bchoices,$btags)

$questions[1] = $bchoices
$answer[1] = arrayfindindex("correct",$btags)

```

Question Text:

```

<style>
  .anstext,
  .answer,
  .mcanswer,
  .mcradio,
  .radiolabel,
  label {
    font-size:14pt !important;
    line-height:1.18 !important;
  }

  input[type="radio"] {
    transform:scale(1.15);
    margin-right:8px;
  }
</style>

<div style="font-family:Arial, sans-serif; font-size:14pt; line-height:1.18; max-width:900px;">
  <div style="font-weight:bold; margin-bottom:8px;">
    Determine whether the relation shown by the graph represents a function. If it is not a function,
    state why not.

```

</div>

<div style="margin:8px 0 12px 0;">
\$graph
</div>

<div style="margin-top:8px;">
 Part A. Does the graph represent a function?
</div>
<div style="margin:4px 0 0 18px;">
 \$answerbox[0]
</div>

///

<div style="margin-top:12px;">
 Part B. Which statement best explains your answer?
</div>
<div style="margin:4px 0 0 18px;">
 \$answerbox[1]
</div>
</div>

Written example:

<div style="font-family: Arial, sans-serif; font-size: 14pt; line-height: 1.15; max-width: 1100px; margin: 0 auto; color: #222;">

 <div style="font-size: 17pt; font-weight: bold; margin: 0 0 6px 0;">Student Guide: Function or Not a Function?</div>

 <div style="border: 1px solid #9bb7d4; background: #f5f9fd; padding: 8px 10px; margin: 0 0 8px 0;">Big idea: A relation is a function when every x-value has exactly one y-value. On a graph, use the vertical line test.</div>

 <table style="width: 100%; border-collapse: collapse; table-layout: fixed;">

 <tbody>

 <tr>

 <td style="width: 63%; vertical-align: top; padding: 0 8px 0 0;">

 <div style="border: 1px solid #999; padding: 8px; margin: 0 0 8px 0;">

 <div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Step-by-Step

Method</div>

 <table style="width: 100%; border-collapse: collapse;">

 <tbody>

 <tr>

 <td style="width: 32px; vertical-align: top; font-weight: bold; color: #1565c0;">1.</td>

 <td style="padding-bottom: 4px;">Pick a vertical line location, which means one x-value.</td>

 </tr>

 <tr>

 <td style="vertical-align: top; font-weight: bold; color: #1565c0;">2.</td>

 <td style="padding-bottom: 4px;">Imagine that vertical line moving across the graph from left to right.</td>

```

        </tr>
        <tr>
            <td style="vertical-align: top; font-weight: bold; color: #1565c0;">3.</td>
            <td>If the vertical line ever hits the graph more than once, the graph is <span
style="color: #9c5a00; font-weight: bold;">not a function</span>. If it only hits once each time, it is
a <span style="color: #2e7d32; font-weight: bold;">function</span>.</td>
        </tr>
    </tbody>
</table>
</div>
<div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Examples of Each Graph
Type</div>
<table style="width: 100%; border-collapse: separate; border-spacing: 6px;">
    <tbody>
        <tr>
            <td style="width: 50%; vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">1. Non-vertical line</div>
                <div style="text-align: center; margin: 3px 0;">$guide_lin</div>
                <div><span style="color: #2e7d32; font-weight: bold;">Yes, function.</span> Each
x-value has one y-value.</div>
            </td>
            <td style="width: 50%; vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">2. Vertical line</div>
                <div style="text-align: center; margin: 3px 0;">$guide_vert</div>
                <div><span style="color: #9c5a00; font-weight: bold;">No, not a function.</span> One
x-value has many y-values.</div>
            </td>
        </tr>
        <tr>
            <td style="vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">3. Vertical parabola</div>
                <div style="text-align: center; margin: 3px 0;">$guide_quad</div>
                <div><span style="color: #2e7d32; font-weight: bold;">Yes, function.</span> A curve
can still pass the test.</div>
            </td>
            <td style="vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">4. Sideways parabola</div>
                <div style="text-align: center; margin: 3px 0;">$guide_side</div>
                <div><span style="color: #9c5a00; font-weight: bold;">No, not a function.</span> Some
vertical lines hit twice.</div>
            </td>
        </tr>
        <tr>
            <td style="vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">5. Exponential curve</div>
                <div style="text-align: center; margin: 3px 0;">$guide_exp</div>
                <div><span style="color: #2e7d32; font-weight: bold;">Yes, function.</span> Each
x-value has one y-value.</div>
            </td>
            <td style="vertical-align: top; border: 1px solid #bbb; padding: 6px;">
                <div style="font-weight: bold;">6. Two exponential branches</div>
                <div style="text-align: center; margin: 3px 0;">$guide_twoexp</div>
            </td>
        </tr>
    </tbody>
</table>

```



```

        <div><span style="color: #9c5a00; font-weight: bold;">No, not a function.</span> Many
x-values have two y-values.</div>
    </td>
</tr>
</tbody>
</table>
</td>
<td style="width: 37%; vertical-align: top; padding: 0 0 0 8px;">
    <div style="border: 1px solid #999; padding: 8px; margin: 0 0 8px 0;">
        <div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">What to Look For</div>
        <div style="margin: 2px 0;"><span style="color: #1565c0; font-weight: bold;">Same
x</span> with exactly one <span style="color: #9c5a00; font-weight: bold;">y</span> &rarr;
function.</div>
            <div style="margin: 2px 0;"><span style="color: #1565c0; font-weight: bold;">Same
x</span> with two or more <span style="color: #9c5a00; font-weight: bold;">y-values</span> &rarr;
not a function.</div>
            <div style="margin: 2px 0;">Straight does not always mean function: a vertical line
fails.</div>
            <div style="margin: 2px 0;">Curved does not automatically mean not a function.</div>
        </div>
        <div style="border: 1px solid #9bb7d4; background: #f5f9fd; padding: 8px; margin: 0 0 8px
0;">
            <div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Shortcut</div>
            <div style="margin: 2px 0;">Use a pencil edge or your finger as a vertical line.</div>
            <div style="margin: 2px 0;">Slide it across the graph.</div>
            <div style="margin: 2px 0;">If it ever touches the graph twice at the same time, answer
<span style="color: #9c5a00; font-weight: bold;">No</span>.</div>
        </div>
        <div style="border: 1px solid #bbb; padding: 8px; margin: 0 0 8px 0;">
            <div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Shortcut Example</div>
            <div style="margin: 2px 0;">A vertical line through a sideways parabola may hit the top
branch and bottom branch.</div>
            <div style="margin: 2px 0;">That means one <span style="color: #1565c0; font-weight:
bold;">x-value</span> has two <span style="color: #9c5a00; font-weight:
bold;">y-values</span>.</div>
            <div style="margin: 2px 0;"><strong>Answer:</strong> <span style="color: #9c5a00;
font-weight: bold;">Not a function.</span></div>
        </div>
        <div style="border: 1px solid #999; padding: 8px; margin: 0 0 8px 0;">
            <div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Common Mistakes</div>
            <div style="margin: 3px 0;"><strong>Mistake:</strong> &ldquo;Curved graphs are not
functions.&rdquo; <br><strong>Fix:</strong> Vertical parabolas and exponential curves can be
functions.</div>
            <div style="margin: 3px 0;"><strong>Mistake:</strong> &ldquo;All straight lines are
functions.&rdquo; <br><strong>Fix:</strong> Vertical lines are not functions.</div>
            <div style="margin: 3px 0;"><strong>Mistake:</strong> &ldquo;Crossing an axis makes it
fail.&rdquo; <br><strong>Fix:</strong> Axis crossings do not decide function or not.</div>
            <div style="margin: 3px 0;"><strong>Mistake:</strong> &ldquo;Having a vertex makes it
fail.&rdquo; <br><strong>Fix:</strong> The direction of the graph matters, not just the
vertex.</div>
        </div>
        <div style="border: 2px solid #2e7d32; padding: 8px; margin: 0;">

```

<div style="font-weight: bold; font-size: 15pt; margin: 0 0 5px 0;">Final Check</div>

<div style="margin: 2px 0;">Part A: Choose Yes if every x has one y.</div>

<div style="margin: 2px 0;">Part A: Choose No if any x has more than one y.</div>

<div style="margin: 2px 0;">Part B should match your reason:</div>

<div style="margin: 2px 0 0 12px;">Yes → “Each x-value has one y-value.”</div>

<div style="margin: 2px 0 0 12px;">No → “Some x-values have two y-values.”</div>

</div>

</td>

</tr>

</tbody>

</table>

</div>